

Revanth krishna Nayani

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SUMMARY

AI/ML Engineer with 4+ years of experience across domains such as AI Platforms, Generative AI, Search & Retrieval Systems, Recommendation Systems, Real-Time Analytics, and Cloud AI Infrastructure. Built and deployed transformer-based applications, distributed inference services, retrieval pipelines, and production ML systems using PyTorch, Ray, Kubernetes, FastAPI, and modern LLM orchestration frameworks. Experienced developing RAG architectures, semantic search platforms, multimodal inference workflows, and GPU-optimized model serving pipelines supporting large-scale enterprise workloads across cloud and edge environments.

SKILLS

Programming Languages: Python, SQL, PySpark, Bash

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, TensorRT, ONNX Runtime

LLM & Generative AI: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LlamaIndex, OpenAI APIs, Claude APIs, Prompt Engineering, AI Agents, Fine-Tuning, LoRA, QLoRA, PEFT, Semantic Search, Embeddings, Hallucination Reduction, Prompt Routing

Vector Databases & Retrieval: Pinecone, Weaviate, FAISS, ChromaDB, Elasticsearch, Hybrid Retrieval, Reranking Pipelines, Context Optimization

MLOps & AI Infrastructure: Kubernetes, Docker, MLflow, Ray, FastAPI, REST APIs, Kubeflow, Airflow, CI/CD, Model Registry, Distributed Inference, GPU Scheduling, AI Observability, Drift Detection, Feature Stores, Model Monitoring

Cloud & Distributed Systems: AWS (SageMaker, EC2, Lambda, EKS, S3, RDS, CloudWatch), Snowflake, Distributed Training, Microservices Architecture

Data Engineering & Streaming: Apache Spark, Kafka, ETL Pipelines, Real-Time Data Processing, Delta Lake, PostgreSQL

Monitoring & Observability: Prometheus, Grafana, Evidently AI, Weights & Biases, LangSmith, OpenTelemetry

EXPERIENCE

NVIDIA

Jun 2024 – Present | Austin, TX

AI/ML Engineer

- Engineered distributed RAG infrastructure using LangChain, LangGraph, vLLM, and Pinecone to support enterprise knowledge retrieval workloads handling 180M+ embedding queries per day with retrieval precision improving from 72% to 91%.
- Deployed GPU-optimized LLM inference services with TensorRT and Kubernetes autoscaling, reducing p95 generation latency by 43% and lowering idle GPU utilization across shared inference clusters.
- Fine-tuned instruction-following LLaMA models using LoRA and QLoRA adapters for internal engineering copilots, improving task completion quality scores by 26% during offline evaluation benchmarks.
- Instrumented AI observability pipelines with Prometheus, Grafana, LangSmith, and OpenTelemetry to trace prompt execution paths, identify retrieval degradation, and reduce unresolved inference incidents by 31%.
- Consolidated multi-stage semantic retrieval workflows through hybrid reranking strategies, adaptive chunking, and contextual compression, cutting vector search response times from 420ms to 130ms under production traffic.
- Hardened multi-agent orchestration services by introducing guardrails, prompt validation layers, and response evaluation frameworks that lowered hallucination frequency by 24% across customer-facing AI workflows.
- Expanded distributed training throughput using Ray-based GPU scheduling and sharded data loaders, shortening large-model fine-tuning cycles by nearly 18 hours per weekly training iteration.

Capgemini

Mar 2020 – Aug 2022 | India

Machine Learning Scientist

- Built fraud detection pipelines using gradient boosting and graph-based anomaly models on 130M+ transaction records, improving high-risk transaction recall from 68% to 86% while reducing false-positive escalation volume by 19%.
- Designed a recommendation engine combining collaborative filtering, embeddings, and session-level behavioral features that increased product engagement CTR by 14% across enterprise retail clients.
- Trained transformer-based NLP models for automated claims and compliance document classification, cutting manual review turnaround time from 11 hours to under 3 hours for internal operations teams.
- Optimized batch inference workflows on Apache Spark and Kafka streams, lowering nightly prediction pipeline runtime by 37% and enabling near real-time feature generation across distributed datasets.
- Automated ML experimentation and model versioning through MLflow, Airflow, and containerized CI/CD pipelines, improving deployment consistency and reducing rollback incidents during production releases.
- Integrated FastAPI inference services with AWS SageMaker endpoints to standardize model serving across forecasting and risk-scoring systems processing over 8M API requests weekly.
- Stabilized real-time recommendation APIs through asynchronous request handling, Redis-backed caching, and adaptive retry policies, decreasing p95 response latency from 540ms to 210ms during traffic spikes.

PROJECTS

AI-Powered Retrieval & Context Intelligence System

- Constructed a semantic search pipeline using hybrid retrieval, vector embeddings, and reranking logic to improve response accuracy across large document datasets, cutting search latency from 460ms to 150ms during high-traffic workloads.
- Delivered distributed inference services using FastAPI, Kubernetes, Ray, and vLLM-based model serving, enabling scalable prompt execution, centralized monitoring, and automated evaluation workflows for production AI applications.

Intelligent Real-Time Traffic Flow Prediction Using Multi-Modal Data

- Engineered multi-agent execution pipelines for task prioritization, document processing, and adaptive workflow routing using LangGraph and asynchronous orchestration services, reducing manual operational effort by 28% across high-volume processing environments.
- Integrated real-time telemetry and monitoring pipelines with Kafka, PostgreSQL, MLflow, and observability tooling to support execution tracking, failure diagnostics, and automated retraining validation for continuously running AI systems.

EDUCATION

Master of Science in Data Science | University of North Texas, Denton, Texas

Aug 2022 – May 2024